SARRTHI: Project Plan and Market Research Notes

Week 2 Deliverable | Reporting period: 13 July to 17 July 2026

Project: SARRTHI, a Proactive Compliance and Payment-Monitoring Platform for MSME Suppliers to Government Transport Bodies

Academic Project — Not an Official Government Portal.

Document Control

Item	Detail
Document	Project Plan and Market Research Notes
Version	1.0
Project week	Week 2 of 14 (13 to 17 July 2026)
Prepared by	Project team: Member 1 (Team Lead), Member 2, Member 3, Member 4
Status	Issued for Team Lead review
Feeds into	Problem Statement and Feasibility Report (Week 3); Requirement Analysis (Week 4); SRS (Weeks 5 and 6)

1. Purpose of this Document

This document has two parts. Part A (Sections 2 to 10) is the project plan: objectives, scope, work breakdown, milestones, roles, technology choices, risks and working practices. Part B (Sections 11 and 12) records the market research and technical feasibility notes that justify the project and shape the requirements phase. Every statistic quoted here is attributed to a named source, in line with the team's Week 2 decision to cite only verifiable figures.

Part A: Project Plan

2. Project Overview

2.1 The problem in brief

Micro, Small and Medium Enterprises (MSMEs) supplying components and fabricated parts to Indian Railways and other government transport bodies are protected by the MSMED Act, 2006. Section 15 limits the payment period to a maximum of 45 days from acceptance of goods, and Section 16 provides for compound interest on delayed amounts. In practice, vendors have no self-serve way to see where each invoice stands against that window. The government's dispute channel, the MSME ODR Portal, activates only after a default has already happened.

2.2 Project objective

To design and build SARRTHI, a web-based management dashboard that gives an MSME supplier early, continuous visibility of its tenders, purchase orders, invoices, payment delays and compliance position, so that problems are seen before they become disputes.

2.3 Proposed solution

SARRTHI is a layered web application: a React and TypeScript front end, a Java Spring Boot REST backend with JWT-based security, and a relational database. It provides a payment clock with interest estimates, compliance intelligence (procurement quota view and buyer risk scoring), a document vault with evidence checklists, and reports with working-capital estimates. It complements the MSME ODR Portal and does not replace it.

3. Objectives and Success Criteria

ID	Objective	Measure of success
O1	Track every invoice against the payment window	Each invoice shows days remaining or days overdue and a status of paid, upcoming or overdue

ID	Objective	Measure of success
O2	Estimate interest on delayed payments	Interest shown for overdue invoices, labelled as an estimate, with its assumptions stated
O3	Provide compliance intelligence	Quota dashboard, rule-based pre-bid risk score and evidence checklist available to the vendor
O4	Manage the procurement lifecycle	Tenders, purchase orders and documents support add, view, edit and delete
O5	Build a secure, testable Java backend	REST API with JWT authentication, role-based access, relational schema and automated unit and integration tests
O6	Document the engineering properly	Problem statement, requirement analysis, SRS v1.0, test summary and final report delivered by 6 October 2026

4. Scope

4.1 In scope for the first version

- User registration, login and business profile (Udyam, GSTIN, PAN).
- Invoice and payment tracking with a configurable 45-day monitoring window and interest estimates.
- Tender tracking, purchase order and delivery management.
- Compliance summary, evidence checklists and buyer risk scores.
- Document management and escalation timeline.
- Dashboard, reports, CSV export and working-capital gap estimate.
- A demo mode with realistic sample data, so the system can be demonstrated without live government data.

4.2 Boundaries and exclusions

- SARRTHI performs no arbitration, conciliation or legal advice. Interest figures are estimates, not guaranteed recoveries.

- No government logos or claimed affiliation. The disclaimer 'Academic Project — Not an Official Government Portal.' is always displayed.
- No live integration with IREPS in the first version, because no public developer API has been identified. The first version relies on vendor-supplied data.
- WhatsApp alerts, Tally and GST e-invoice integration, collective intelligence and native mobile apps are deferred to later phases.

5. Work Breakdown Structure and Schedule

The project runs for 14 weeks, from 6 July to 6 October 2026.

WBS	Phase	Weeks	Key deliverables
1	Initiation	1	Project brief, work breakdown structure, repository, tool setup
2	Research and planning	1 to 3	Market research, project plan, feature comparison, Problem Statement and Feasibility Report
3	Requirements	4 to 6	Module-wise requirement analysis, use cases, priority matrix, SRS draft and completion
4	UI/UX design	6	Sitemap, user flows, wireframes, design brief
5	Foundations	7	Frontend skeleton, backend structure, shared components, utilities, demo data layer
6	Core development	8 to 11	Backend modules, all application pages, payment logic, reports, sign-in and bindings
7	Integration and testing	12	Frontend connected to backend, UI and backend test suites, defect list
8	Stabilisation and closure	13 to 14	Defect fixing, responsive and accessibility checks, SRS v1.0, final verification and project report

6. Milestones

No.	Milestone	Target date
M1	Project kick-off complete, tools and repository ready	10 July 2026
M2	Project plan and market research notes (this document)	17 July 2026
M3	Problem Statement and Feasibility Report	24 July 2026
M4	Requirement analysis and use cases	31 July 2026
M5	SRS first draft (Sections 1 to 5)	7 August 2026
M6	SRS complete and UI/UX design delivered	14 August 2026
M7	Foundations ready (shell, components, utilities, demo data)	21 August 2026
M8	First backend modules and first pages	28 August 2026
M9	All planned pages developed	11 September 2026
M10	Metrics, bindings, sign-in and refinement	18 September 2026
M11	Integration and testing complete	25 September 2026
M12	SRS v1.0 issued	1 October 2026
M13	Final verification and project report	6 October 2026

7. Team Roles and Responsibilities

Member	Primary responsibilities
Member 1 (Team Lead)	Planning and timeline, repository and architecture decisions, user identity and access, application shell and routing, integration and sign-in
Member 2	Invoice and payment monitoring, the 45-day window and interest rules, date and currency utilities

Member	Primary responsibilities
Member 3	Compliance intelligence, reports and dashboard, visual direction and UI consistency
Member 4	Tenders, purchase orders and documents, data models and demo data, backend design note, integration testing

All members contribute to the documentation. Names replace 'Member 1 to 4' in the final submission.

8. Technology Plan and Decision Log

Layer	Choice	Reason
Backend	Java with Spring Boot (Web, Data JPA, Security)	Java is the mandated backend language; Spring Boot gives a clean layered structure and mature security support
Persistence	Spring Data JPA with Hibernate; PostgreSQL (H2 for local development)	ACID-compliant relational storage suits financial records
Security	Spring Security with JWT, role-based access	Stateless authentication suited to a REST API
API documentation	OpenAPI / Swagger	Self-documenting endpoints for testing and review
Frontend	React with TypeScript, Vite, Tailwind CSS	Agreed in Week 1; type safety and fast iteration
Testing	JUnit and Mockito (backend); Vitest (frontend)	Standard tools for each stack
Version control	Git and GitHub	Shared repository with branch-based work

Decision	Outcome	Week
D1	Use React with TypeScript for the frontend	1
D2	Use Java (Spring Boot) for the backend	1 to 2
D3	First version works on vendor-supplied data; no dependency on government APIs	2
D4	Position SARRTHI as a complement to the MSME ODR Portal, not a replacement	2
D5	Cite only verifiable sources; record the year of every statistic	2

9. Risk Register

ID	Risk	Likelihood	Impact	Mitigation
R1	No public API for IREPS, so government data is unavailable	High	High	Build on vendor-supplied data first; treat government data as a later enhancement
R2	Statistics come from different years and sources	High	Medium	Cite only verifiable sources, record the snapshot year, and avoid adding figures from different years
R3	System is mistaken for legal advice or an official portal	Medium	High	Always show the disclaimer; label interest as an estimate; state calculation assumptions
R4	Scope grows across nine feature modules	High	Medium	Phase the features; fix the first version around the payment clock and interest calculator
R5	Small team and fixed deadline	Medium	High	Clear module ownership, weekly reports, early integration

ID	Risk	Likelihood	Impact	Mitigation
R6	Frontend and backend drift apart	Medium	Medium	Agree data types early and generate bindings from the backend interface
R7	Personal and financial data handling	Medium	High	Follow the principles of India's DPDP Act, 2023: consent, minimisation, secure storage
R8	Merge conflicts in shared files	Medium	Low	Branch per member, frequent merges, review before merging

10. Communication, Version Control and Quality Plan

- **Reporting:** each member submits weekly progress to the Team Lead, who compiles the weekly report. Issues and next-week plans are recorded every week.
- **Shared folder:** all documents are stored in one shared folder, with the latest version named and dated.
- **Version control:** one Git repository, a branch per member, changes merged after review.
- **Definition of done:** a feature is done when it works end to end, has a passing test where logic is involved, handles empty, loading and error states, and follows the design brief.
- **Quality gates:** type-check, lint and the test suites must pass before each integration.

Part B: Market Research Notes

11. Market Research Notes

11.1 Market context

Metric	Figure	Source
Delayed payments owed to MSMEs, FY24 (inflation-adjusted)	₹7.34 lakh crore	GAME, FISME and C2FO, November 2025
MSME dues reported locked in overdues annually	₹11 lakh crore	Rajya Sabha reply, 6 February 2023
Samadhaan backlog (historical snapshot)	₹21,108 crore across 90,000+ applications	MSME Samadhaan portal data
Samadhaan applications with no response from the buyer	14,739 of 18,203 (about 81%)	Industry report citing Samadhaan data
Active MSME sellers on GeM	68,286 sellers; ₹20,263 crore order value	Press Information Bureau, GeM statistics

These figures come from different years and are not additive. They show the scale and persistence of the problem and are not a single reconciled total. Each must be re-checked against its primary source before the final report. Government sources name Railways, electricity and aeronautics as the sectors owing MSMEs the most.

11.2 Statutory framework

Provision	What it provides	Relevance to SARRTHI
MSMED Act, Section 15	Payment within the agreed time, which may not exceed 45 days from acceptance; where nothing is agreed in writing, payment is due within 15 days	Basis of the payment clock

Provision	What it provides	Relevance to SARRTHI
MSMED Act, Section 16	Compound interest with monthly rests at three times the RBI bank rate on delayed amounts	Basis of the interest estimate
MSMED Act, Section 18	Reference of disputes to the MSEFC (Facilitation Council)	Escalation path that SARRTHI prepares for but does not perform
Income Tax Act, Section 43B(h)	A buyer cannot claim a deduction for delayed MSME payments until they are actually paid	Gives buyers a reason to pay on time
Public Procurement Policy for MSEs, Order 2012 (as amended)	Central Ministries, Departments and CPSEs must procure at least 25% of annual purchases from MSEs, including 4% from SC/ST-owned and 3% from women-owned MSEs	Basis of the quota dashboard
MSME classification, S.O. 1364(E), effective 1 April 2025	Micro up to ₹2.5 crore investment and ₹10 crore turnover; Small up to ₹25 crore and ₹100 crore; Medium up to ₹125 crore and ₹500 crore	Used for eligibility checks

Correction recorded: early research notes quoted a 20% procurement quota. The current minimum under the amended Public Procurement Policy Order is 25%. SARRTHI uses 25% as its default and keeps it configurable.

11.3 Existing solutions

Solution	Purpose	Limitation relative to the gap
MSME Samadhaan Portal (2017 to 2025)	Filing and tracking of delayed-payment complaints; now an archive for earlier cases	Reactive; no new filings since 15 October 2025
MSME ODR Portal (from 27 June 2025)	Online negotiation, then MSEFC conciliation and arbitration; built-in interest calculator	Starts only after a default; Micro and Small only; no live payment clock
Government e-Marketplace (GeM)	Government buying and selling platform	Procurement tool, not a payment-compliance monitor
IREPS	Indian Railways e-procurement system	No public vendor-facing API found; not a monitoring tool
TReDS platforms (for example RXIL, M1xchange, Invoicemart)	Invoice discounting for early payment	Financing only; does not track the legal window
General accounting software	Ledgers and receivable ageing	Not built around the MSMED window, quota or buyer risk
Tender consultancies	Tender discovery and bid preparation	Pre-award only; excludes post-award payment visibility

11.4 Feature comparison

Feature	MSME ODR Portal	General accounting software	SARRTHI (proposed)
Live 45-day payment clock before a default	No	Limited (ageing only)	Yes

Feature	MSME ODR Portal	General accounting software	SARRTHI (proposed)
Continuous interest estimate	At filing only	Limited	Yes
Procurement quota monitoring	No	No	Yes
Buyer payment-risk view	National figures only	No	Yes, per buyer
Evidence checklist and document vault	At filing only	Limited	Yes
Working-capital gap estimate	No	Limited	Yes
Covers Medium enterprises	No	Yes	Yes
Legally binding outcome	Yes	No	No (feeds into the ODR process)

Entries reflect the public descriptions reviewed in Weeks 1 and 2. 'Limited' means partly addressed.

11.5 Target users

User	Needs	Pain points
MSME owner	One view of money owed, due dates and risk	Learns of a delay only after cash is locked up; unsure how to claim interest
Accounts or office staff	Fast entry of orders, invoices and payments	Manual tracking in spreadsheets; scattered evidence
Administrator (future)	Verification and support tools	No central view of data quality

Medium enterprises are excluded from the ODR and MSEFC mechanism entirely, so they may be an especially motivated early-adopter group.

11.6 Positioning and value proposition

SARRTHI works upstream of the dispute process. It tells a vendor what is due, what is late, what it may be owed and what evidence is still missing, so that the vendor can act early. The positioning is complementary: SARRTHI prepares information and never performs legal functions.

11.7 Research quality notes

- Statistics from different years are labelled with their year and are never added together.
- Qualitative claims rely on documented, named cases and not on anecdotes.
- Any figure that cannot be verified from a primary source is marked as unverified.

12. Technical Feasibility Notes

Area	Finding	Decision
Backend platform	Spring Boot gives layered structure, security, data access and testing in one stack	Adopt Java with Spring Boot
Data access	No public IREPS API identified	Use vendor-supplied data; keep government data as a later option
OCR for scanned invoices	Tesseract (through a Java wrapper) or a cloud OCR service are both workable	Defer to a later phase; manual entry first
WhatsApp alerts	Requires an approved business account and message templates	Defer; in-app alerts first, email as a fallback
Data protection	The DPDP Act, 2023 requires consent, purpose limitation and the ability to delete data	Design for consent and deletion from the start
TReDS	Existing platforms already provide invoice discounting	Offer a referral, not an integration
Hosting	Containerised deployment keeps the choice of cloud open	Docker, with PostgreSQL in production

Area	Finding	Decision
Team skills	Team is comfortable with Java and TypeScript	Proceed as planned

13. Next Steps (Week 3)

- Write the Problem Statement and Feasibility Report (Abstract, Introduction, Problem Statement, Proposed System, Feasibility and Scope).
- Re-verify every statistic in Section 11.1 against its primary source.
- Finalise the feature-comparison table for the report.
- Start the 2025 ODR Portal review in more detail.

References

1. Government of India, Ministry of MSME. Micro, Small and Medium Enterprises Development Act, 2006, Sections 15, 16, 18, 20 and 21.
2. Ministry of MSME. Notification S.O. 1364(E), 21 March 2025 (revised MSME classification).
3. Ministry of MSME and Development Commissioner (MSME). Public Procurement Policy for Micro and Small Enterprises (MSEs) Order, 2012, and its FAQs.
4. Global Alliance for Mass Entrepreneurship, FISME and C2FO. MSMEs' Access to Finance and Timely Payments. November 2025.
5. Rajya Sabha Unstarred Question No. 427, answered 6 February 2023.
6. MSME Samadhaan Portal and MSME ODR Portal (odr.msme.gov.in), Ministry of MSME.
7. Government e-Marketplace statistics, Press Information Bureau.
8. Income Tax Act, 1961, Section 43B(h), inserted by the Finance Act, 2023.
9. Digital Personal Data Protection Act, 2023.